How it fits together

Threat intelligence → detections for Defender XDR, Google SecOps, Splunk and Elastic

AUTOMATED MULTI-VENDOR THREAT HUNTING & DETECTION SUITE

open source · self-hosted · works without AI by default · @cmdaltr

core flow · colour = input (Hunt / Analyse)
optional AI (off unless enabled)
config, context and influence
coming soon
needs user interaction

INTERFACES

- **Web UI** FastAPI, server-rendered
- **archer-cli** start / stop / restart supervisor
- **Console API**
- **MCP server**

INPUTS — EVERY RUN

- **Company profile** — Domain → sector, geography, observed tech stack
- recommends groups
- **Hunt** — Threat group or alias · CVE(s) · ATT&CK technique IDs
- **Analyse** — Report, blog URL, or STIX/TAXII/MISP → techniques and IOCs
- optional
- **Research** — Threat Feed: the week's research from ~26 vendor and government sources, tagged by industry, nation and motive. Watchlists flag new posts (optional email).
  - ▸ Your stack filter

GENERATION PIPELINE

- **06 IOC sweeps** branch — Report and CVE indicators, with expiry
- IOCs
- **OPTIONAL AI** · off unless enabled — Ask Archer · six agents: repair, ask, triage, sweep, audit, tune · LLM via local Ollama or a cloud provider
- **01 Procedure retrieval** — Group → techniques → procedures
- **02 Extraction, hygiene + stack filter** — Placeholders, generic words and known-good removed; skips what you don't run
- **03 Query builders** — Defender KQL · SecOps YARA-L · Splunk · Elastic
- **04 Validate + save** — Conditions that parse but can't fire are removed; the rest join the shared YAML library
- **05 Correlation engine** — Chains detections across tactics on one host or account in a time window
- tech stack
- feeds the pipeline · checks YAML indicators

OUTPUTS

- **Results** — Threat briefing (coverage and gaps) · detections per SIEM · correlations · attack flow (drag-and-drop) · STIX export · detections-as-code · IOC flow · validation checklist
- **Library** — Saved detections and correlations
- **Coverage** — ATT&CK matrix
- **Flows** — Detectable vs not, per attack flow
- **Detections as code · Assessments** — git export + run diff · red/blue test outcomes (CSV, Navigator)

context

CONFIG — ONE-TIME SETUP

- **Profile** — SIEM platforms · Threat Feed industries · API keys
- **bespoke/ mappings** — Org-specific index names and overrides, kept out of git
- **Technology stack** — Platforms and products you run

EMULATE

- Which logs record each technique (native OS logs first, Sysmon and EDR optional), by technique, group or data source · indicators checked against the recordings · Defender XDR's built-in alerting per technique

TOOLS

- **Regex tester** — RE2 and PCRE2 run in their real engines
- **MITRESaw** — ATT&CK procedures and their indicators
- **Cyber Powers map** — National cyber capability on a world map

BACKEND · INTELLIGENCE SOURCES context, not inputs

storage: SQLite

- **ATT&CK + MITRESaw** — STIX bundle · MITRESaw (git submodule): procedure extraction across direct, campaign and software attribution, with an evidence workbook
- **NVD + public PoCs** — CVE research and exploit indicators
- **OSINT rule index** — Elastic, Splunk, Sentinel and SecOps public rules, plus their documented false positives
- **LOLBAS**
- **LOLDrivers**
- **LOLRMM**
- **Known-good lists** — First-party domains · DNS resolvers · empty-file hashes
- **Recorded attack telemetry** — Splunk attack_data · OTRF Security-Datasets · EVTX-to-MITRE-Attack · macOS-ATTACK-DATASET
- **Microsoft GUIDE** — Defender XDR's own alerting per technique across 5,000+ organisations, with triage grades
- **Your lab** — COMING SOON — Atomic Red Team runs on your test VMs, with Defender for Endpoint's rows
- **Test VMs** — COMING SOON — Windows · Linux · macOS → record-atomics

archer-cli — the same pipeline from a terminal

one command: archer-cli · console modules · python -m maintenance jobs

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core flow · colour = input (Hunt / Analyse)
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config, context and influence
coming soon
dimmed = web UI only
needs user interaction

>_ INTERFACES

archer-cli start|stop|restart|run|status|logs · serve · clear-cache · install · docker
--port --host --reload/--no-reload --ollama/--no-ollama --ollama-models DIR --skip-index
logs [archer|ollama|supervisor] -f -n

archer-cli console
use <module> · set/unset <OPTION> · show options|modules
info · search · run · back
ARCHER_URL · ARCHER_TOKEN (or ARCHER_EMAIL / ARCHER_PASSWORD)

archer-cli preview <group>
--siem mde,secops,splunk,elastic --index/-i NAME
--limit N --output/-o FILE

archer-cli mcp
MCP server: hunt_* · wait_for_run · get_run_results
save_query_fix · prompts
repair/ask/triage/sweep/audit/tune

archer-cli offline generate|correlate|hunt (emulate: coming soon) — the pipeline without the server · generate --correlate --llm/--no-llm --hypothesis · correlate --min-queries --max-queries --yaml-dir --output

INPUTS — EVERY RUN

Company profile
use intel/company DOMAIN
INDUSTRY GEOGRAPHY TECH_STACK
SUPPLY_CHAIN GROUPS (top N|names|0)

GROUPS

Hunt
use hunt/group GROUP · hunt/cve CVE
use hunt/technique TECHNIQUES
EXCLUDE (secops,mde,splunk,elastic)
MERGE_BY_TACTIC MERGE_BY_PARENT BRIEFING

Analyse
use intel/report URL | FILE
MODE all|iocs|review TECHNIQUES USE_AI

optional

Research
use intel/feed SEARCH SHOW relevant|all
INDUSTRIES REFRESH WAIT

use research/actors KEYWORDS TOP
MODE AND|OR SCOPE ACTORS EXPORT xlsx

GENERATION PIPELINE

01 Procedure retrieval
GROUP · HYPOTHESIS

02 Extraction, hygiene + stack filter
LOL default on · OSINT
OSINT_COMPARE
LOL = LOLBAS/LOLDrivers/LOLRMM
stack: from profile (no option)

IOCs

06 IOC sweeps branch
intel/report MODE iocs · hunt/cve

03 Query builders
SIEM / EXCLUDE · INDEX

04 Validate + save
VALIDATE Docker SIEMs
python -m archer.contract validate|lint|report
DRY_RUN stdout only
writes data/groups/*.yaml

05 Correlation engine
CORRELATE MIN_RELEVANCE
MIN_QUERIES MAX_QUERIES

OPTIONAL AI · off unless enabled
MODE llm|fallback|standard · PROVIDER assigned|none|anthropic|openai|google|deepseek|qwen|grok
MODEL · MAX_CALLS
use ai/ask · ai/briefing GROUP TOP_N · agent/repair RUN SCOPE LIMIT
agent/ask · agent/triage URL · agent/sweep · agent/audit · agent/tune · agent/history SESSION CONTINUE
APPROVE ask|all

WAIT · OUTPUT <zip> — every server-driven module

tech stack

feeds the pipeline · checks YAML indicators

OUTPUTS

Results
use results RUN LATEST QUERIES OUTPUT <zip>

Library WEB UI ONLY
Saved detections and correlations

Coverage WEB UI ONLY
ATT&CK matrix

Flows WEB UI ONLY
Detectable vs not, per attack flow

Detections as code · Assessments WEB UI ONLY
git export + run diff · red/blue test outcomes (CSV, Navigator)

context

CONFIG — ONE-TIME SETUP

Profile
use ai/settings ASK_PROVIDER ASK_MODEL
DETECTION_PROVIDER DETECTION_MODEL
MAX_CALLS LOCAL_ENDPOINT LOCAL_MODEL
REMEMBER

bespoke/ mappings
files only · gitignored
bespoke/mappings · prompts · query_templates

Technology stack
use intel/company TECH_STACK
otherwise web UI only

EMULATE

python -m archer.telemetry --build
[--source splunk|otrf|evtx|macos|lab]
python -m archer.telemetry --guide <zip|folder|csv>
--mde-export <run dir>

TOOLS

Regex tester WEB UI ONLY

MITRESaw python -m archer.mitre.procedures [--full] [-o FILE]
incremental by default

Cyber Powers map WEB UI ONLY

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storage: SQLite

ATT&CK + MITRESaw
STIX bundle · MITRESaw (git submodule): procedure extraction, evidence workbook
python -m archer.mitre.procedures

NVD + public PoCs
CVE research and exploit indicators
fetched on demand

OSINT rule index
Elastic, Splunk, Sentinel, SecOps public rules + documented false positives
scripts/data/update_osint.py
--force --no-index --elastic --splunk
--azure-sentinel --chronicle

LOLBAS LOLDrivers LOLRMM
fetched by update_osint.py

Known-good lists
First-party domains · DNS resolvers · empty-file hashes
fetched on demand

Recorded attack telemetry
Splunk attack_data · OTRF Security-Datasets · EVTX-to-MITRE-Attack · macOS-ATTACK-DATASET
python -m archer.telemetry --build

Microsoft GUIDE
Defender XDR's own alerting per technique, 5,000+ organisations
python -m archer.telemetry --guide

Technique yield
python -m archer.build.technique_yield
--rebuild · --reset-llm [TID…]

Your lab
COMING SOON
Atomic Red Team runs on your test VMs, with Defender for Endpoint's rows

Test VMs
COMING SOON
Windows · Linux · macOS · record-atomics